FPGA-Based Magnetotelluric Data Acquisition and DSP System for Landslide Detection

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***Abstract*—Landslides are among the most dangerous natural hazards, damaging infrastructure, environment, and human life. This paper presents an FPGA-based data acquisition and DSP system for landslide detection using the magnetotelluric (MT) method, processing natural EM field variations in real time. The architecture (Xilinx Artix-7) includes EM sensors, signal conditioning, a 10-bit 100 MSPS ADC, buffering, 2048-point FFT computation, magnitude estimation, and UART communication to detect abnormal EM variations linked to landslide activity. Validated through behavioural simulation and post-implementation analysis in Vivado, the design achieves high-speed operation, efficient resource utilisation, and an estimated on-chip power of 0.333 W from Vivado post-implementation netlist analysis—making it a scalable platform for future landslide early-warning applications.**

***Index Terms*—FPGA, Landslide Detection, Magnetotelluric (MT), Data Acquisition System, Digital Signal Processing (DSP), Fast Fourier Transform (FFT), Real-Time Monitoring, Artix-7 FPGA.**

I. Introduction

Landslides cause significant loss of life and infrastructure damage, particularly in tectonically active regions such as the Himalayan foothills. Reliable early warning requires continuous, real-time monitoring of subsurface changes preceding slope failure. The Magnetotelluric (MT) method is a passive electromagnetic technique that characterises subsurface resistivity without artificial source infrastructure [1], [2], making it well suited for landslide monitoring.

MT records five field components (E_x, E_y, H_x, H_y, H_z) over 10^{-3} Hz to 10 kHz, with depth sensitivity $\delta \approx 503\sqrt{\rho/f}$ (m), enabling investigation of both shallow slip surfaces and deeper structures. Li et al. [3] presented an EM acquisition circuit for landslide monitoring but lacked a complete FPGA-based real-time DSP architecture. Liu et al. [4] proposed an FPGA-based multi-channel system with wireless telemetry, limited to low bandwidth and sampling rates. Ren et al. [5] introduced a lightweight Zynq SoC-based acquisition module lacking an integrated high-speed MT processing pipeline. A need therefore remains for a low-power, high-speed FPGA platform integrating real-time MT acquisition with on-chip DSP for continuous landslide monitoring.

This paper presents an FPGA-based MT acquisition and DSP system for landslide detection on a Xilinx Artix-7 (Digilent Nexys 4 DDR), offering low-power, high-speed, real-time on-chip processing suited for portable, long-term monitoring. The system performs entirely on-chip:

- 10-bit ADC capture at 100 MSPS (single-channel);
- 2048-point pipelined radix-2 FFT for real-time spectral analysis;
- CORDIC-based magnitude computation $|X[k]| = \sqrt{\mathrm{Re}^2[k] + \mathrm{Im}^2[k]}$;
- Trigger-based acquisition with adjustable gain and threshold;
- UART interface for host-PC transfer and configuration;
- Estimated on-chip power of 0.333 W (Vivado post-implementation netlist analysis), enabling energy-efficient continuous monitoring.

Section II presents the theoretical background of the Magnetotelluric (MT) method. Section III details the proposed FPGA architecture and its sub-modules. Section IV presents the simulation, hardware implementation, and post-implementation results for on-chip FPGA. Section V concludes the paper and outlines directions for future work. Finally, section VI provides the acknowledgment of the financial support received for this research.

II. Magnetotelluric Method: Theoretical Background

The MT method is a passive EM technique using natural electric and magnetic fields to infer subsurface resistivity [1], [2]. Fields above 1 Hz arise from thunderstorm activity, below 1 Hz from solar wind–magnetosphere interaction [6], [7]. The field decays exponentially with depth via the skin depth [8]:

$$\delta = 503\sqrt{\frac{\rho}{f}} \quad [\mathrm{m}], \tag{1}$$

where ρ (Ω·m) is resistivity and f (Hz) is frequency, giving depth sensitivity from metres to tens of kilometres [6], [7]. The impedance tensor $\mathbf{Z}(\omega)$ relates orthogonal field components [1], [2]:

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}. \tag{2}$$

Apparent resistivity and phase are:

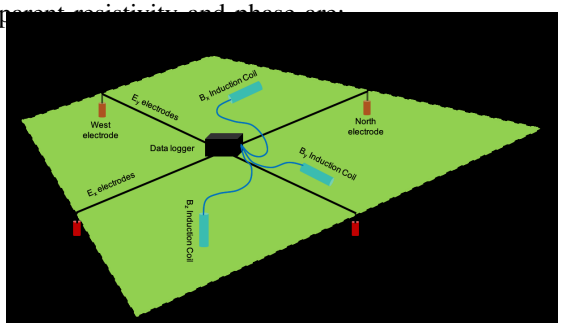


Fig. 1. Schematic of a five-component MT field deployment showing orthogonal electric dipoles (E_x, E_y), three-axis induction coil magnetometers (B_x, B_y, B_z), and a central data logger at a single survey site [7].

$$\rho_{a,ij} = \frac{1}{\mu_0\,\omega}\left|Z_{ij}\right|^2, \tag{3}$$

$$\phi_{ij} = \arctan\!\left(\frac{\mathrm{Im}[Z_{ij}]}{\mathrm{Re}[Z_{ij}]}\right), \tag{4}$$

Table I compares the principal EM methods.

TABLE I
COMPARISON OF EM METHODS FOR LANDSLIDE MONITORING

Method	Freq. Range	Max Depth	Key Limitation
MT [9]	0.001 Hz–10 kHz	100 km	Cultural EM noise
CSAMT [6]	0.1 Hz–100 kHz	1 km	10–30 kW transmitter
ERT [10]	DC–10 Hz	50 m	Shallow; galvanic contact
GPR [11]	10 MHz–3 GHz	10 m	Severe clay attenuation
TEM [8]	0.001 Hz–100 kHz	500 m	Large transmitter loop

III. Proposed FPGA-Based Acquisition Architecture

A. System Overview

The proposed system is implemented on a **Xilinx Artix-7** FPGA (Digilent Nexys 4 DDR) for on-chip MT-based landslide monitoring [3]: a 10-bit ADC input at 100 MSPS feeds a 2048-point real-time FFT, and spectral magnitude is transmitted to a host PC via UART [4].

As shown in Fig. 2, the ADC input is sampled, buffered, and processed by the FFT core, with resulting magnitudes sent via UART for real-time display. The signal chain is:

$$\underbrace{\text{ADC}}_{100\text{M},10b} \xrightarrow{\texttt{rd_adc}} \underbrace{\text{RAM}}_{2048} \xrightarrow{\texttt{dec}} \underbrace{\text{FFT}}_{2048\text{pt}} \xrightarrow{\text{Re,Im}} \underbrace{|X(k)|}_{\text{Sq,Sum,Rt}} \xrightarrow{\texttt{UART}} \text{PC} \tag{5}$$

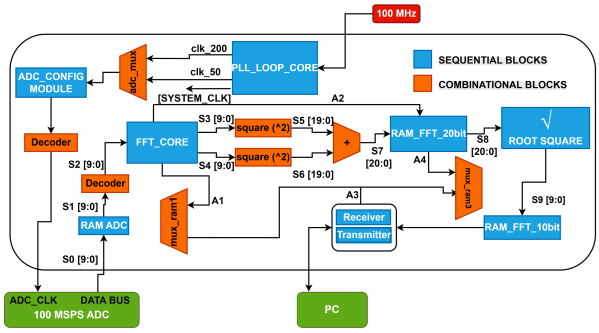


Fig. 2. Top-level block diagram of the proposed FPGA-based data acquisition and DSP system.

B. Sub-Module Descriptions

The design is partitioned into nine sub-modules: `PLL_loop` generates 50/100/200 MHz clocks; `read_adc` captures 10-bit ADC samples; `RAM_ADC` buffers 2048 samples; `decoder` converts binary-offset to two's complement; `FFT` performs a 2048-point pipelined radix-2 transform; `Square/Sum` and `Root` compute the magnitude, stored in `RAM_FFT`; `serial_tx/rx` handles UART communication; and `TOP` provides FSM control and integration.

C. Finite State Machine Control

Three co-operating FSMs in the TOP module govern the workflow. **FSM 1**, on command `'P'`, sequences: *Wait* → *Trig* → *Acq* → *FFT* → *FFT_write* → *Square* → *Send* → *Wait*; commands `A`–`D` set time-base, trigger level, slope, and gain in *Wait*. **FSM 2**, on the 100 MHz ADC clock, fills `RAM_ADC` with 2048 samples during *Acq*, else computes $\Delta x = x(k+1) - x(k)$ for trigger detection. **FSM 3** transmits the FFT magnitude bytes, sync bytes (`'F','F','T'`), and raw ADC samples via UART.

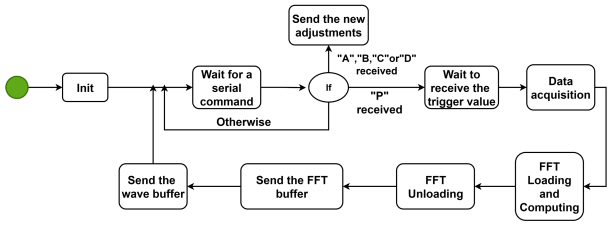


Fig. 3. FSM 1: signal-processing state machine. States `s1`–`s4` perform $s(k)$ calculation, ADC acquisition, $s(k+1)$ computation, and differential $x(k+1) - x(k)$ compilation.

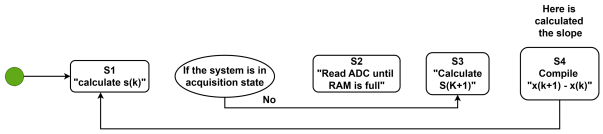


Fig. 4. FSM 2: UART transmission state machine sequentially dispatching the FFT buffer, flag bytes `F`, period value `T`, and waveform buffer, conditioned on the TX-enable flag.

D. FFT Engine and Magnitude Computation

The 2048-point pipelined radix-2 Cooley–Tukey FFT [4] runs at 200 MHz, giving 48.828 kHz resolution (Δf =

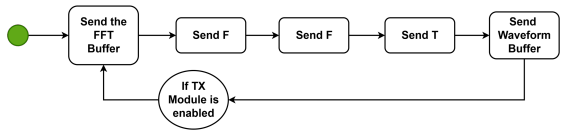


Fig. 5. FSM 3: top-level control cycling through initialisation, command reception, trigger, acquisition, FFT processing, and transmission, with a reconfiguration branch.

$f_s/N = 100\times10^6/2048$) and 62.51 μs latency, with ADC samples passing through the `decoder` before the FFT core. Spectral magnitude is:

$$|X[k]| = \sqrt{\mathrm{Re}[k]^2 + \mathrm{Im}[k]^2}, \tag{6}$$

computed via `Square`, `Sum`, and CORDIC-based `Root` modules, yielding a 10-bit magnitude stored in `RAM_FFT_10bit`. The 48.828 kHz/bin resolution is used for broadband spectral analysis to detect abnormal EM variations. For resolving low-frequency MT signals below 10 kHz, decimation-based sub-band processing is planned as future work.

E. Key System Parameters

The system runs on a Xilinx Artix-7 (XC7A100T, Nexys 4 DDR) FPGA with a single-channel 10-bit SAR differential ADC at 100 MSPS (50 MHz Nyquist). The 2048-point radix-2 pipelined FFT runs at 200 MHz, giving 62.51 μs latency and 48.828 kHz resolution per bin. The current hardware acquires one MT field component at a time; for full five-component MT acquisition (E_x, E_y, H_x, H_y, H_z), five such channels can be instantiated by replicating the same RTL design with Clock Domain Crossing (CDC) handling, planned as future hardware revision. Post-implementation: 50 DSP slices (20.83%), 7.5 Block RAMs (5.56%), 0.333 W power, with UART to the host PC.

IV. Simulation and Post-Implementation Results

A. Simulation Environment

Functional correctness was verified via behavioural simulation in Vivado 2025 using two Verilog testbenches. The primary testbench (`tb_TOP`) simulates the end-to-end pipeline using a 64-entry sine wave LUT as ADC stimulus, with key parameters in Table II. Field measurements and real landslide datasets are not part of this work; sensor integration and field trials are planned as future work.

TABLE II
Key Simulation Parameters of `tb_TOP`

Parameter	Value	Description
Clock period	10 ns	100 MHz system clock
Timescale	1 ns/1 ps	Time unit/precision
ADC word width	10 bit	Matches physical ADC
Sine LUT size	64 entries	One complete period
UART trigger byte	`0x50`	ASCII `'P'`
Startup settling	100 cycles	PLL lock emulation
Sim. timeout	5,000,000	Deadlock detection

The testbench bypasses UART timing by injecting the trigger byte directly via Verilog `force`/`release`, cutting latency from ∼8 000 to 6 cycles. It is self-checking, passing only if the FSM returns to `wait_state` after cycle 1000, confirming deadlock-free completion.

B. FSM State Transition Verification

The TOP module FSM was verified through all seven states in simulation: `wait_state` (0 ns, idle/polling UART) → `trig_state` (1,075,000 ns, on `'P'`) → `acq_state` (1,305,000 ns, ADC acquisition) → `fft_state` (FFT activated) → `fft_write` (bins written to RAM) → `square_state` (magnitude computed) → `send_state` (UART transmission).

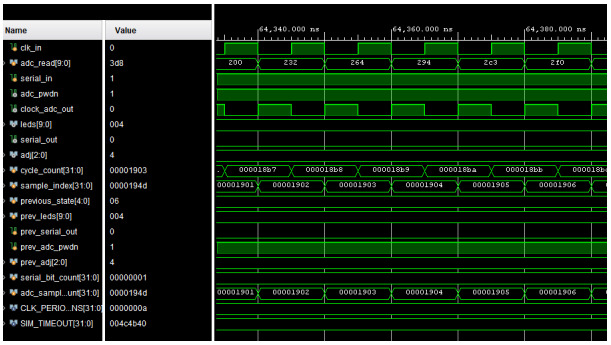


Fig. 6. Behavioural simulation waveform of the top-level FPGA design (`tb_TOP`) in Vivado.

C. FFT Core Simulation

The second testbench (`tb_fft_vivado2025`) validated the 2048-point FFT core using multi-tone inputs applied one frequency at a time, confirming correct spectral analysis for each of the five MT field component types (E_x, E_y, H_x, H_y, H_z) [4], achieving 62.51 μs latency and 48.828 kHz resolution at 200 MHz. Fig. 6 verifies complete system operation (ADC acquisition, UART communication, FSM control), while Fig. 7 confirms correct pipelined FFT operation with output coefficients generated after the initial data-loading interval.

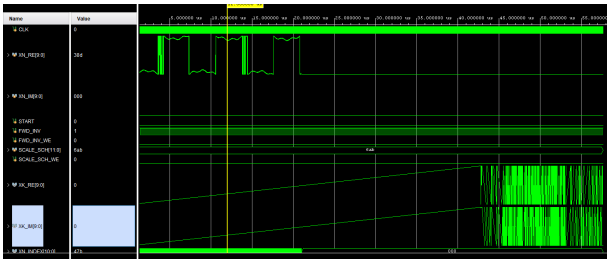


Fig. 7. Behavioural simulation waveform of the 2048-point FFT core in Vivado.

D. Post-Implementation Resource Utilisation

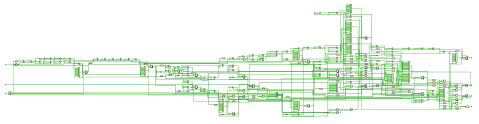


Fig. 8. RTL schematic of the complete FPGA design generated by Vivado.

TABLE III
FPGA RESOURCE UTILISATION SUMMARY (XILINX ARTIX-7 XC7A100T, VIVADO POST-IMPLEMENTATION NETLIST)

Module	Slice LUTs	DSPs	BRAMs
TOP (full)	3067 (4.84 %)	50 (20.83 %)	7.5 (5.56 %)
FFT core	1524	48	6.0
RAM_ADC	602	0	0
RAM_FFT	601	0	0
Root_Square	155	0	0

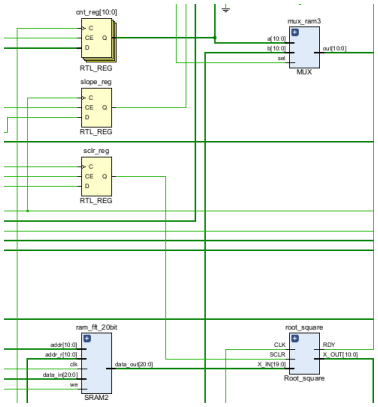


Fig. 9. Partial RTL schematic of the FFT magnitude computation module.

E. Power Analysis

The on-chip power of 0.333 W was obtained from Vivado post-implementation netlist analysis of the Artix-7 XC7A100T FPGA only; this is not a direct hardware measurement. The MMCM clock generation block dominates at 0.096 W (41% of dynamic power), summarised in Table IV.

TABLE IV
ON-CHIP POWER BREAKDOWN (VIVADO POST-IMPLEMENTATION NETLIST ANALYSIS, ARTIX-7 XC7A100T)

Component	Power (W)	Share (%)
Dynamic Power – 0.234 W (70 %)		
MMCM	0.096	41
DSP	0.031	13
BRAM	0.025	11
Signals	0.024	10
Clocks	0.023	10
I/O	0.021	8
Logic	0.015	7
Static Power – 0.098 W (30 %)		
Device static	0.098	30
Total	**0.333**	**100**

F. Comparative Performance

Table V compares the proposed system against published FPGA-based geophysical DAQ systems on the basis of FPGA platform, ADC resolution, channel count, sampling rate, and on-chip power. Power values of Shu [12], Ren [5], and Liu [4] are all NA; therefore no direct power comparison is claimed.

TABLE V
COMPARISON OF FPGA-BASED GEOPHYSICAL DAQ SYSTEMS (CH. = CHANNEL COUNT; NA = NOT AVAILABLE IN CITED WORK)

Reference	FPGA	ADC	Ch.	Rate	Power
Shu [12]	Intel Cyclone	12-bit	3	80 MSPS	NA
Ren [5]	Zynq-7020	16-bit	4	100 kSPS	NA
Liu [4]	Cyclone V	24-bit	5	10 kSPS	NA
Proposed	**Artix-7**	**10-bit**	**1**	**100 MSPS**	**0.333 W***

*Vivado post-implementation netlist estimate; not a hardware measurement.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented an FPGA-based MT DAQ and DSP system on Xilinx Artix-7 (Nexys 4 DDR), validated via behavioural simulation in Vivado 2025, achieving 3067 LUTs (4.84%), 50 DSPs (20.83%), 7.5 BRAMs (5.56%), and 0.333 W estimated on-chip power. With a 100 MSPS 10-bit ADC, 2048-point pipelined FFT, CORDIC magnitude computation, and seven-state FSM, it offers the highest sampling rate in the comparison group, suited for portable landslide monitoring in regions such as the Barak Valley.

B. Future Work

Future work includes **sensor integration and field trials**, **five-channel hardware expansion** with CDC for simultaneous E_x, E_y, H_x, H_y, H_z acquisition, **decimation-based subband processing** for MT signals below 10 kHz, **Zynq SoC migration** (ARM Cortex-A9 for GPS timestamping and network management), **on-chip anomaly detection** (lightweight FFT-based classifiers for pre-landslide EM signatures), and **hardware power measurement** as part of prototype validation.

VI. ACKNOWLEDGEMENT:

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